

Optimize

Health

Performance

Longevity

Acheivement



BrainAvatar™ 3D Brain Imaging, QEEG, and Neurofeedback Systems



Atlantis
Series

Discovery
Series



Freedom
Series

 **BrainMaster**
TECHNOLOGIES
Mastering the Brain, Changing Lives

What is Neurotherapy?

- Also called Neurofeedback (NFB), EEG biofeedback, brainwave training, or brain fitness
- Involves placing small sensors (similar to tiny microphones) on the head to record brainwave activity using real-time displays of electroencephalography (EEG)
- Information gathered during the process can be used to teach the brain to operate more efficiently by providing visual and auditory feedback, utilizing a reward incentive to the client's brain as their brainwave patterns improve and self-regulation occurs
- Can include a QEEG assessment (quantitative electroencephalography), where your client's brain function is compared to a normative database, and converted to a brain map to be used as a tool for validation and guidance
- Usually involves 2-3 sessions/week, fifteen to forty minutes per session. Number of sessions depend upon client goals

Neurotherapy is a process that includes:

- observing brainwave activity and function, giving information related to why and how a client may struggle with certain issues
- creating a training and learning program to better target underlying neurophysiological factors that may exacerbate symptoms or hinder progress
- assisting the brain to improve or change brainwave patterns by incorporating a positive reward incentive to better improve function using visual and auditory feedback, such as music, games, imaging, and even educational or entertainment movies
- working with the client as functioning improves, and providing emotional support and cognitive tools that help the client complete the process of improvement

How can Neurotherapy help my clients?

NFB can be a favorable option for all ages with a wide range of concerns. It is also oftentimes used for healthy brains looking to reach peak performance. For many patients seeking this form of training, their brains have become stuck in a rigid pattern of responding. NFB training can help the brain become more flexible and efficient.

In addition, many clients report:

- improved quality of sleep
- uplifted mood, emotional balance
- reduction in overall stress
- a change in symptoms
- improved peak performance (athletes, musicians, executives, students)
- feeling empowered through being an active participant

Why is Neurotherapy important to me as a practitioner?

In the rapidly developing field of brain health, an understanding of brain function and how it affects the mental, emotional, and physical states of individuals seeking help is becoming critical to the quality of services being provided. Neurotherapy offers evidence-based options demonstrated to be effective over the last fifty years.

- non-toxic, non-invasive alternative to medication
- assist in gathering information about why an individual may be having symptoms based on brainwave activity
- target underlying dysregulation in brain activity
- gather valuable baseline brain data
- promote positive changes at the source of the problem
- track changes in brain function (validation)

What do I charge my clients? Does insurance pay?

National average*:

- Neurofeedback Training \$75-\$150 per session
- QEEG/Brain Mapping \$450-\$500 per pre/post assessment

*Insurance reimbursement for brain mapping, assessment, and biofeedback varies greatly from state to state and is dependent upon your licensure. Please check your state's regulations for more information. Common codes are available upon request.

What educational certification or training opportunities are available?

- Workshops
- In-Services
- Webcasts/Video Tutorials
- Mentoring
- Comprehensive NFB certification (BCIA, QEEG, etc.)
- Free, unlimited Tech Support for our affiliates

For more information, contact www.StressTherapySolutions.com





The neurophysiology and science are compelling.

Some practitioners are unconvinced or very cautious as they look into Neurotherapy. However, once they attend a very good course that makes the neurophysiology understandable, it can be an “a-ha” moment. (Note: brainwave training is a form of neuroscience and takes some effort to understand.) When a patient changes his/her EEG, the patient is by definition changing activation and timing patterns. These patterns affect many pathways and feedback loops in the brain. When you start to understand such mechanisms, Neurotherapy becomes undeniable as an effective alternative treatment or adjunct to your current treatment modality.

Symptoms

Seeing someone experience a change rapidly that cannot be explained any other way never fails to amaze. The brain can shift states very quickly. As an example, in many training courses we’ve observed, clients reported clear, significant changes in state after several days of training. Changes oftentimes influence sleep, mood, alertness, focus, or attention. For some clients, changes — often quite unexpected changes — occur reasonably soon.

Experienced practitioners quickly note that these unexpected changes cannot be simply placebo, and can only be explained due to Neurotherapy training. Comparing to a normative database, although not a necessity, is even more compelling, as changes are confirmed and documented.

NOTE: Approximately ten or more training sessions, depending on the severity of the client’s condition, are usually needed before any short-term, transitory changes should be expected to hold.

Research

Though “lack of research” or “limitations” are often cited, that response usually comes from people who haven’t read the publications or much of the research. When practitioners read enough of the research and review outcome studies, it’s very hard to dismiss. Neurofeedback is considered evidence-based.

Practitioners will often look into Neurotherapy after hearing another practitioner talk about its success and impact. Providers who are experienced in Neurofeedback can offer some excellent clinical insights. In addition, many practitioners enter this field because someone they have personally met has experienced a success that was simply too life-changing to dismiss.



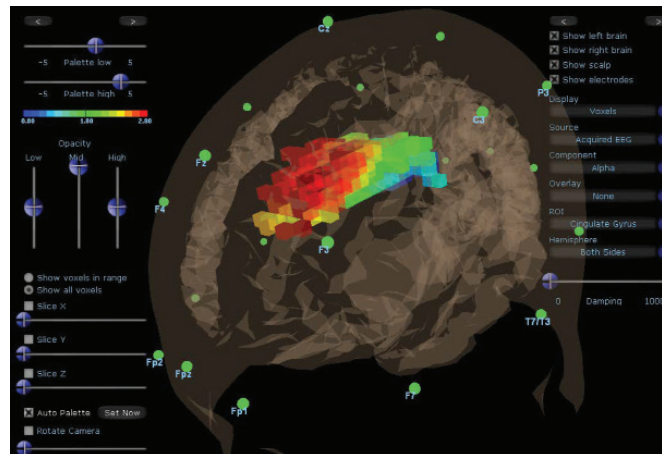
What if you could see...

Introducing the most sophisticated, state-of-the-art, real-time imaging system available.
A white paper by Thomas F. Collura BrainMaster Technologies, Inc. & Brain Enrichment Center.

Live sLORETA Brain Imaging & Biofeedback using a Normative Database

Introduction

A new method for brain imaging was constructed by combining the sLORETA localization algorithm with a reference database. Using this approach, it is possible to reconstruct brain electrical activity in real-time, and produce accurate 3D images of brain activation in a range of EEG component bands. All 6,239 voxels of brain activity are computed in real-time (8 frames per second) in 8 frequency bands, and converted into Z-Scores reflecting the instantaneous current-source density compared to a normative population.

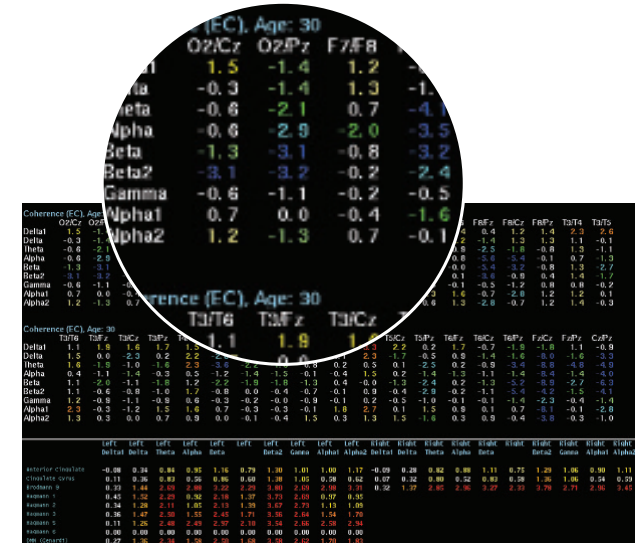


Above: A live reconstruction and imaging using sLORETA. It depicts training within the Cingulate Gyrus.

Methodology

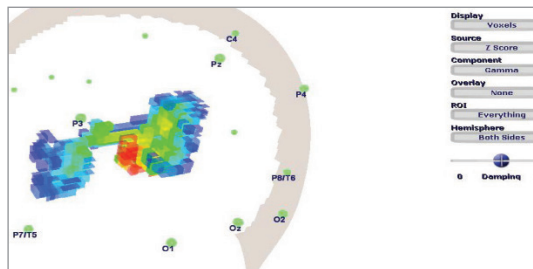
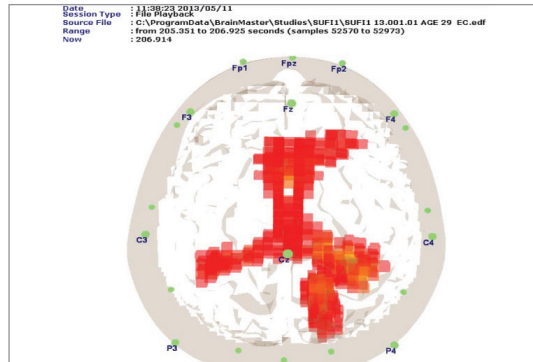
The sLORETA algorithm was described by Pascual-Marqui (2002), and the normative reference is based on the database first described by John, et al. (2000). The examples that follow demonstrate its dynamic functionality.

Specific results are shown demonstrating the ability of this technique to isolate brain functional regions including networks and hubs, as a function of instantaneous brain state.



1. Individual State Patterns

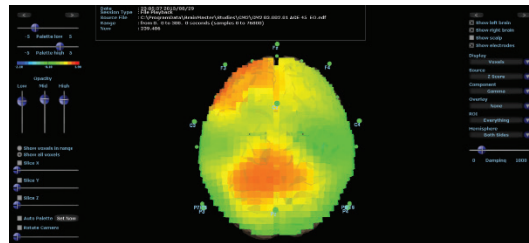
In one case of extreme pain control, a hub of awareness was detected at the -3.0 Z-Score level.



In this image of gamma activity, a profound excess is found in the mesiotemporal regions, corresponding to an internalized state.

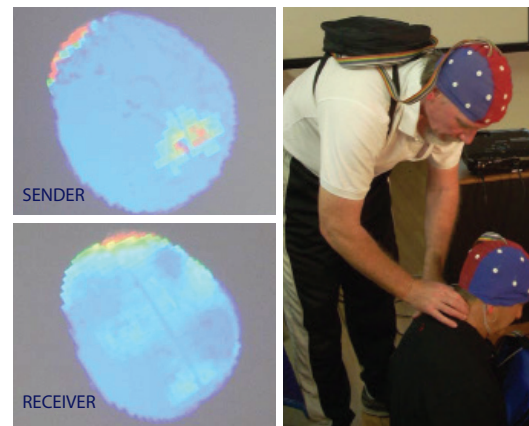
2. Isolation of Brain Functional Regions

In other cases of extreme performance, individual patterns reflecting states of awareness, caution, and emotional tone were recognized.



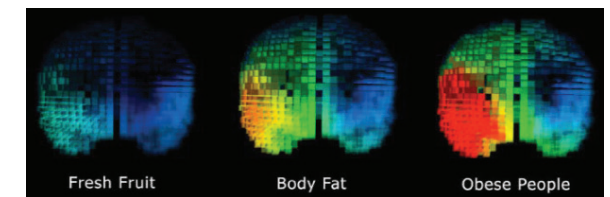
3. Complementary Activation — Sender & Receiver

Another example, in a case of energy healing, showed a unique pattern of complementary activation between the sender and receiver.

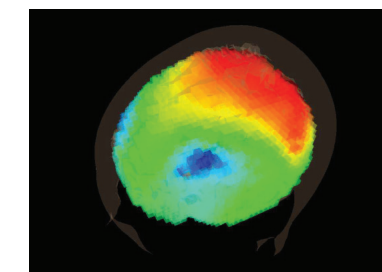


4. Ipsative Responses to Cues

The BrainAvatar™ process examines the asymmetry in the prefrontal cortex and identifies gamma wave bursts to assess the underlying subconscious responses at the very moment of decision-making. This process provides scientific evidence that an evoked, emotionally-laden response results in corresponding brain activity, and exposes the match to self-reported explanations. The process documents both the intensity of human emotional response, as well as the directionality of the response.



Above: Dominant gamma wave activity in the left prefrontal cortex signifies acceptance, and increased activity in the right represents avoidance (Bonstetter, et al., 2012a, Bonstetter, et al., 2012b).



Left: Focal deficit in Broca's Area in an elderly individual with a speech deficiency.

REFERENCES

- John, E. R., Pritchep, L.S. & Eaton, P. (1987). Normative data banks and neurometrics: Basic concepts, methods and results or norm construction. In A. Remond (ed.), Handbook of electroencephalography and clinical neurophysiology: Vol. III. Computer analysis of the EEG and other neurophysiological signals (pp. 449-495). Amsterdam: Elsevier.
- Pascual-Marqui, R.D. (2002). Standardized low resolution Brain electromagnetic tomography sLORETA: technical details. Methods & Findings in Experimental & Clinical Pharmacology. 24D:5-12
- Bonstetter, R., Collura, T. F., Bonstetter, B. J. & Hebets, D., (2012a). Trait and state-based depression indicators measured using sLORETA imaging and quantification. Presented at the 2012 meeting of the International Pharmacology-EEG Group (IPEG), New York, New York, November, 2012.
- Bonstetter, R., Collura, T. F., Hebets, D., & Bonstetter, B. J. (2012b). Uncovering the belief behind the action. Neuroconnections, Winter 2012, 20-23
- General Motors, Inc. (2013). 2014 Corvette Stingray Biometric Test Drive. Accessed from <http://vimeo.com/channels/corvette>.
- Peper, E. & Hall, H. (2013) Sufi self-piercing with EEG monitoring demonstration. Presented at the 2013 meeting of the Association for Applied Psychophysiology and Biofeedback, March 15, Portland, Oregon.

Which System is Right for My Clinic?

With our broad selection of hardware and software, you can train as few as one or two channels or as many as 19 channels of EEG. With options like original Live Z-Score Training, multimedia player for DVDs*, games, photic stimulation, and visual and audio feedback, MicroTesla (PEMF), 3D Brain Imaging, and general biofeedback choices — it's a no-brainer that BrainMaster is your single-source provider to enhance your practice and impact the lives of your patients!



Atlantis Series

Neurofeedback Therapy Systems

The perfect start up system for acquisition, real-time assessment and therapy

Options include:

- up to 4 channels of EEG and 4 channels of AUX
- photic, vibratory, and auditory feedback
- database driven Z-Score Training
- self-adjusting protocols
- multimedia player with dimmer screen
- clinical use or practitioner-supervised remote (home) EEG



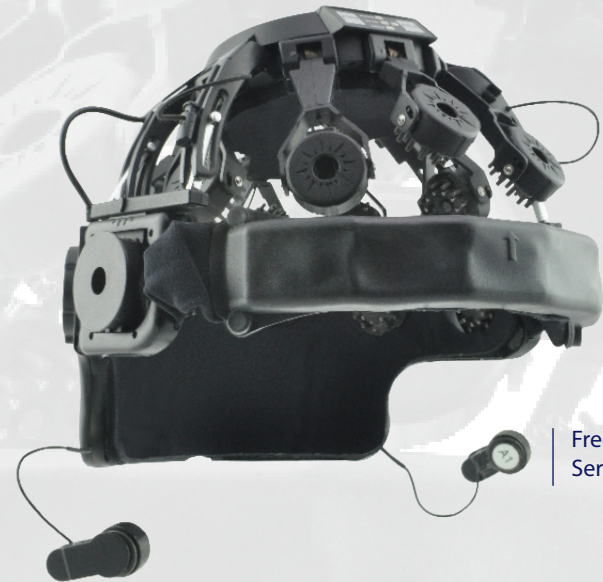
Discovery Series

QEEG, 3D Imaging, and Neurofeedback Therapy All in One

Increase your capabilities by utilizing the most advanced technology available in Neurofeedback today. Assess, image, and train up to 19 channels simultaneously. Live sLORETA Projection and more, all in an easy to use, automated system that takes the guesswork out of it for you.

Options include:

- up to 2 channels of AUX for additional biofeedback
- multimedia player with dimmer screen
- live, 3D imaging of the brain
- compare your results to various databases
- see the brain in real time
- isolate individual regions, hubs, networks, Brodmann areas, and lobes



Freedom Series

State-of-the-Art Dry Technology

Take it to the next level! All the same benefits as the Discovery Series in a no gel/no paste application.

- research-grade
- dry wireless headset
- rapid application of 21 sensors
- Self-donned and ready to record EEG in under 5 minutes
- Comfortable, portable design suitable for light ambulation in office or lab environment

Freedom Series is a third-party product and labeled per original manufacturer policy for research purposes.

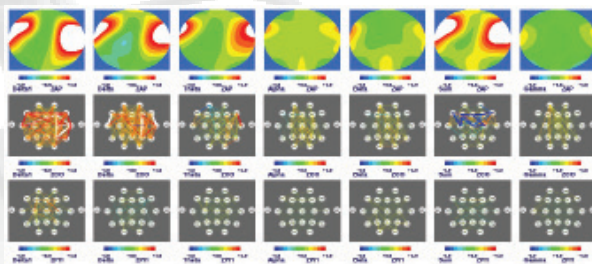
Remote User Systems

Allow your client to train their own brain! BrainMaster makes it easy for you to impact as many people as possible by offering a portable, easy-to-use, high-quality, take-home unit for those that would benefit from several sessions a week, and are unable to make it in to the office as often as they would like, due to schedule/logistics. We offer a complete, turn-key package that includes 3 or more devices and everything you need to get started. We can even recommend a business model tailored to fit your needs.

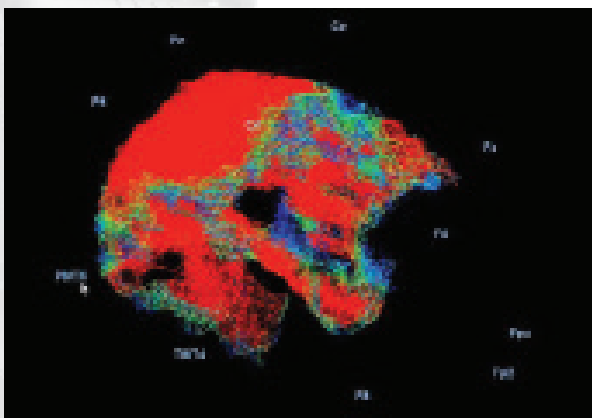
ASK ABOUT LEASE TO OWN OPTIONS

Please call today for more information! 440.232.6000

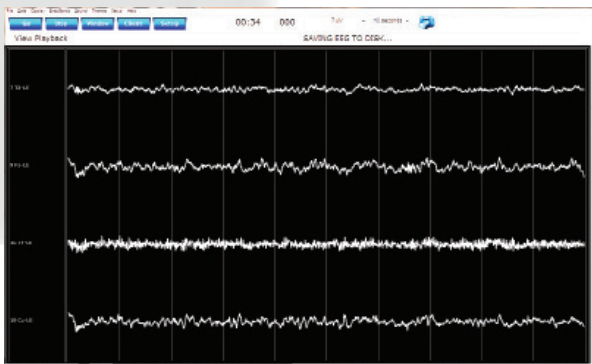
BrainMaster invented the original Live Z-Score method 2-19 Channels. All devices and software are sold to or on the order of a licensed practitioner. *All DVDs must be obtained from an authorized distributor.



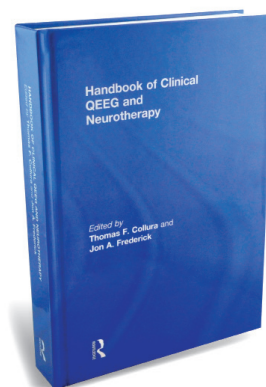
Sample 2D qEEG Surface Maps



sLORETA 3-D Head Display



Raw EEG



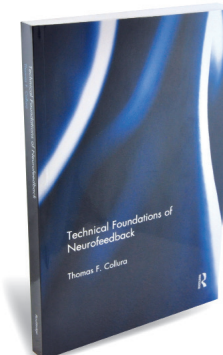
Edited by Thomas Collura
and Jon A. Frederick

Handbook of Clinical QEEG and Neurotherapy

This book is an essential resource describing a wide range of approaches and technologies in the areas of quantitative EEG (QEEG) and Neurotherapy, including Neurofeedback and Neuromodulation. It emphasizes practical, clinically useful methods, reported by experienced clinicians who have developed and used these approaches first hand. These chapters describe how the authors approach and use their particular combinations of technology, and how clients are evaluated and treated. This resource, which is encyclopedic in scope, provides a valuable and broad, yet sufficiently detailed account to help clinicians guide the future directions in client assessment and neurotherapeutic treatment. Each contribution includes literature citations, practical information related to clinical interventions, and clinical outcome information.



Order Hardcopy or
Kindle using this
QR CODE



by Thomas Collura
PhD, QEEG-D,
BCN, LPC

Technical Foundations of Neurofeedback

Technical Foundations of Neurofeedback provides, for the first time, an authoritative and complete account of the scientific and technical basis of EEG biofeedback. Beginning with the physiological origins of EEG rhythms, Collura describes the basis of measuring brain activity from the scalp and how brain rhythms reflect key brain regulatory processes. He then develops the theory, as well as the practice of, measuring, processing, and feeding back brain activity information for biofeedback training. Combining both a "top-down" and a "bottom-up" approach, Collura describes the core scientific principles as well as current clinical experience and practical aspects of Neurofeedback assessment and treatment therapy. Whether the reader has a technical need to understand Neurofeedback, is a current or future Neurofeedback practitioner, or only wants to understand the scientific basis of this important new field, this concise and authoritative book will be a key source of information.



Order Hardcopy or
Kindle using this
QR CODE

A blue-tinted photograph of children running in a park, serving as the background for the top half of the page.

Mastering the Brain, Changing Lives

Add Neurofeedback to Your Clinic

Zero down, deferred payments for 90 days
36, 48, and 60 month lease to own

BrainMaster Technologies prides itself on more than 23 years of service. Our products are manufactured in the USA and are FDA 510K cleared as both clinical EEG and biomedical devices. We offer world-class tech support and access to some of the best educational opportunities in the industry. Partner with us, and help us change lives, one brain at a time!



BrainMaster Technologies, Inc.
195 Willis Street Bedford, OH 44146
440.232.6000

BrainMaster.com | Sales@brainm.com

All devices and software are sold to or on the order of a licensed practitioner to be used under scope of practice and competence within state licensure for EEG Biofeedback, stress reduction or relaxation purposes. BrainAvatar, sLORETA, database guidance products are labeled under the category of EEG BIOFEEDBACK per industry standards. QEEG related software use is for assessment purposes only and not diagnostic. Freedom series is a third-party product and labeled per original manufacturer policy for research purposes. All insurance matters are between the practitioner and the insurance company only.